

REPUBLIC OF RWANDA

EASTERN PROVINCE

KIREHE DISTRICT

KIGINA SECOTOR

KIREHE ADVENTIST TECHNICAL SECONDARY SCHOOL

SUBJECT: DEVOPS

72 Linux Commands and Their Uses from the DevOps Techniques Application Manual

System Information Commands

1. **uname -r** - Shows the Linux kernel version.
2. **uname -m** - Displays the machine hardware architecture.
3. **uname -a** - Displays all available system information (kernel name, version, hardware, etc.).

CPU Management Commands

4. **lscpu** - Displays detailed information about the CPU architecture.
5. **cat /proc/cpuinfo** - Displays processor details directly from the /proc file system.
6. **nproc** - Shows the number of processing units (cores) on the system.
7. **mpstat** - Displays CPU usage statistics per processor (requires sysstat package).

Memory Management Commands

8. **free -h** - Displays the total amount of free and used memory (RAM and swap).
9. **vmstat** - Reports virtual memory statistics including processes, paging, and CPU activity.
10. **top** - Provides a real-time dynamic view of system processes and memory consumption.

Disk and Storage Commands

11. **df -h** - Displays disk space usage for file systems.
12. **lsblk** - Lists information about available block devices (disks and partitions).
13. **du -sh <directory>** - Displays the disk usage of a specific directory.
14. **mount** - Lists all mounted file systems.
15. **umount <device>** - Unmounts a mounted device.

Network Information Commands

- 16. **ifconfig** - Displays network interface configurations.
- 17. **ip a** - Shows detailed IP address configurations.
- 18. **ping <host>** - Tests network connectivity to a specified host.
- 19. **traceroute <host>** - Displays the route and measures transit delays of packets across a network.
- 20. **netstat -tuln** - Lists all active listening ports and network statistics.
- 21. **ss -tuln** - Displays active TCP/UDP connections.

Hardware Information Commands

- 22. **sudo dmidecode** - Displays detailed system hardware information (requires root privileges).
- 23. **lshw** - Provides detailed hardware configuration information.
- 24. **lsusb** - Lists USB devices connected to the system.
- 25. **lspci** - Lists all PCI devices.

System Uptime and Load Commands

- 26. **uptime** - Displays system uptime, current time, number of logged-in users, and load averages.
- 27. **top** - Also provides information on system uptime, processes, and resource usage.
- 28. **htop** - An interactive process viewer (requires installation).

File and Directory Management Commands

- 29. **ls** - Lists files and directories in the current directory.
- 30. **mkdir <directory_name>** - Creates a new directory.
- 31. **rm <file_name>** - Removes a file; use **-r** to remove directories.
- 32. **cp <source> <destination>** - Copies files and directories.
- 33. **pwd** - Displays the current working directory path.
- 34. **touch <file_name>** - Creates an empty file.
- 35. **mv <source> <destination>** - Moves or renames files and directories.

Text Processing Commands

- 36. **cat <file_name>** - Displays the content of a file.
- 37. **grep <pattern> <file_name>** - Searches for a pattern within a file.
- 38. **sed 's/old/new/g' <file_name>** - Replaces occurrences of a string within a file.
- 39. **awk '{print \$1}' <file>** - Extracts and processes text data.
- 40. **sort <file>** - Sorts lines in text files.
- 41. **uniq <file>** - Filters out repeated lines.

Process Management Commands

- 42. **ps** - Displays a list of currently running processes.
- 43. **kill <process_id>** - Terminates a process using its process ID.
- 44. **top** - Monitors real-time system processes and resource usage.
- 45. **htop** - Displays interactive real-time process monitoring.
- 46. **nice -n <priority> <command>** - Runs a process with a specified priority.
- 47. **renice <priority> -p <PID>** - Changes the priority of an already running process.

Package Management Commands

- 48. **apt-get install <package_name>** - Installs packages on Debian-based systems.
- 49. **yum remove <package_name>** - Removes packages on Red Hat-based systems.
- 50. **apt-get update** - Updates the list of available packages.
- 51. **dpkg -i <package.deb>** - Installs a Debian package file.
- 52. **rpm -ivh <package.rpm>** - Installs an RPM package.

User and Group Management Commands

- 53. **useradd <username>** - Creates a new user.
- 54. **passwd <username>** - Sets or changes a user's password.
- 55. **groupadd <group_name>** - Creates a new group.
- 56. **users** - Displays currently logged-in users.
- 57. **id <username>** - Displays user ID and group ID information.
- 58. **chown <owner>:<group> <file>** - Changes file ownership.
- 59. **chmod 755 <file>** - Changes file permissions.

System Control Commands

- 60. **systemctl start <service_name>** - Starts a service.
- 61. **systemctl stop <service_name>** - Stops a service.
- 62. **systemctl restart <service_name>** - Restarts a service.
- 63. **shutdown now** - Immediately shuts down the system.
- 64. **reboot** - Reboots the system.
- 65. **journalctl** - Views system logs.

Miscellaneous Commands

- 66. **cd <directory>** - Changes the current working directory.
- 67. **sudo** - Executes commands with superuser (root) privileges.
- 68. **alias ll='ls -aF'** - Creates command shortcuts (aliases).
- 69. **history** - Displays the command history.
- 70. **date** - Displays or sets the system date and time.
- 71. **whoami** - Displays the current logged-in user.
- 72. **hostname** - Displays or sets the system hostname.

